

PIF

2026

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1TPIFI

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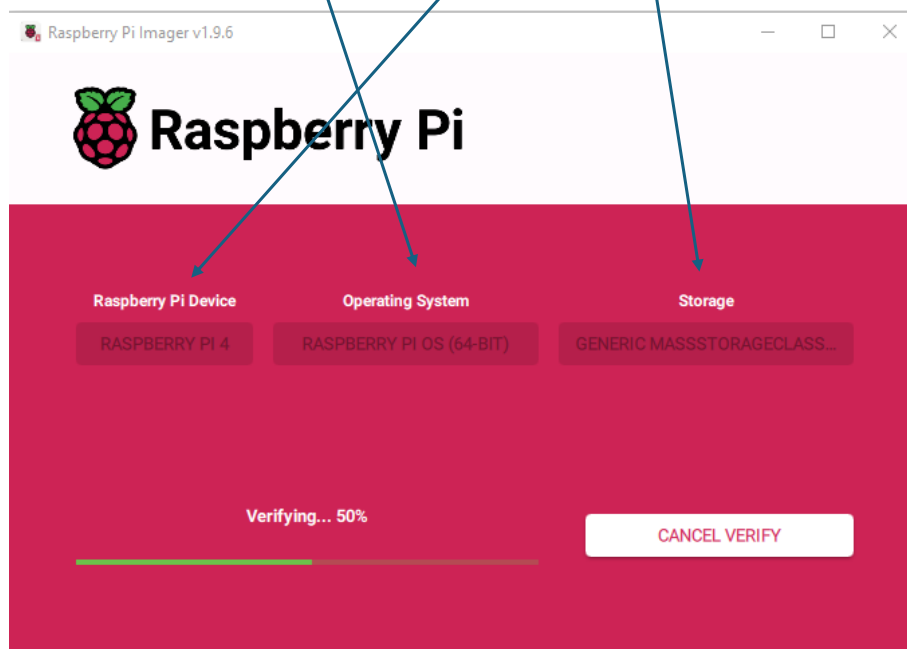
Lab sprints

L1 Raspberry Pi

Raspberry Pi OS

First you need to download the Raspberry Pi OS and flash it on a SD card.

1. Install the Raspberry Pi imager: <https://www.raspberrypi.com/software>
2. Once opened you can specify your Raspberry Pi device (for us Raspberry Pi 4 Model B), the operating system you want and the device you want to save it on (in our case a SD card).



3. Once you chose everything press write and you will be asked if you want to flash the SD card or if you want to Edit the settings, press edit settings.

- On the general tab you can set a hostname, create your username and password, connect to a WLAN and select your timezone and keyboard layout.



General

Services

Options

☒ Set hostname:

GilJa879

.local

☒ Set username and password

Username:

gilja879

Password:

••••••••••••••••

☒ Configure wireless LAN

SSID:

LPEM_B13

Password:

••••••••

☒ Hidden SSID

Wireless LAN country:

LU

▼

☒ Set locale settings

Time zone:

Europe/Luxembourg

▼

Keyboard layout:

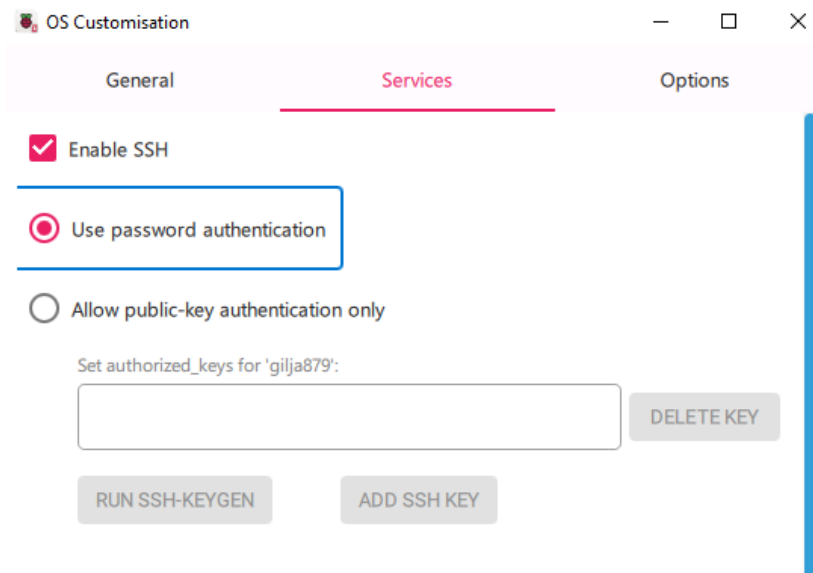
ch

▼

CANCEL

SAVE

5. On the Services tab, enable SSH and for now you can use a password since we don't have a public-key yet.



OS Customisation

General Services Options

☒ Enable SSH

☒ Use password authentication

☐ Allow public-key authentication only

Set authorized_keys for 'gilja879':

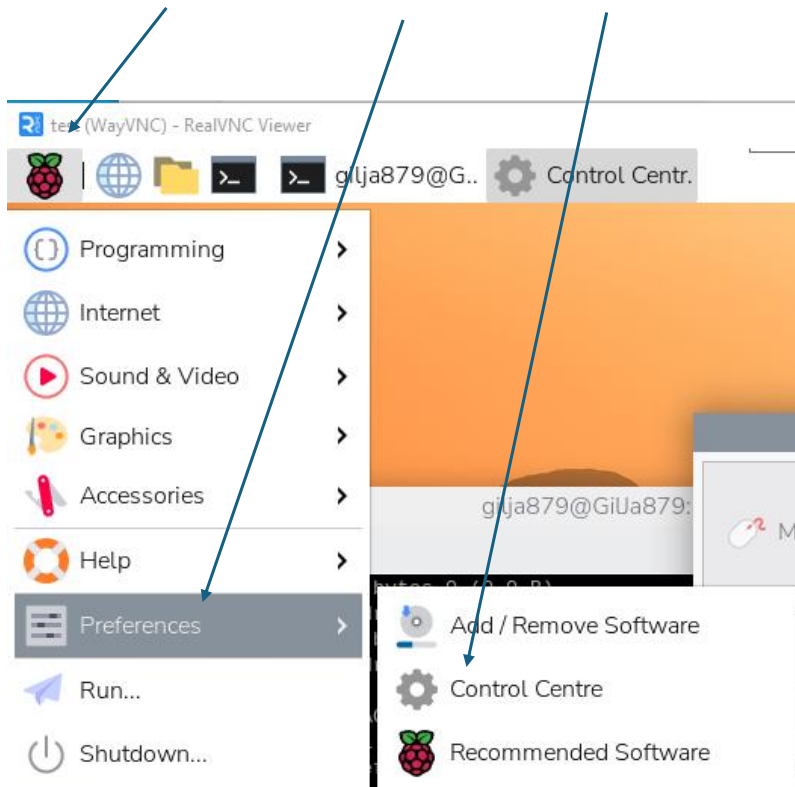
DELETE KEY

RUN SSH-KEYGEN ADD SSH KEY

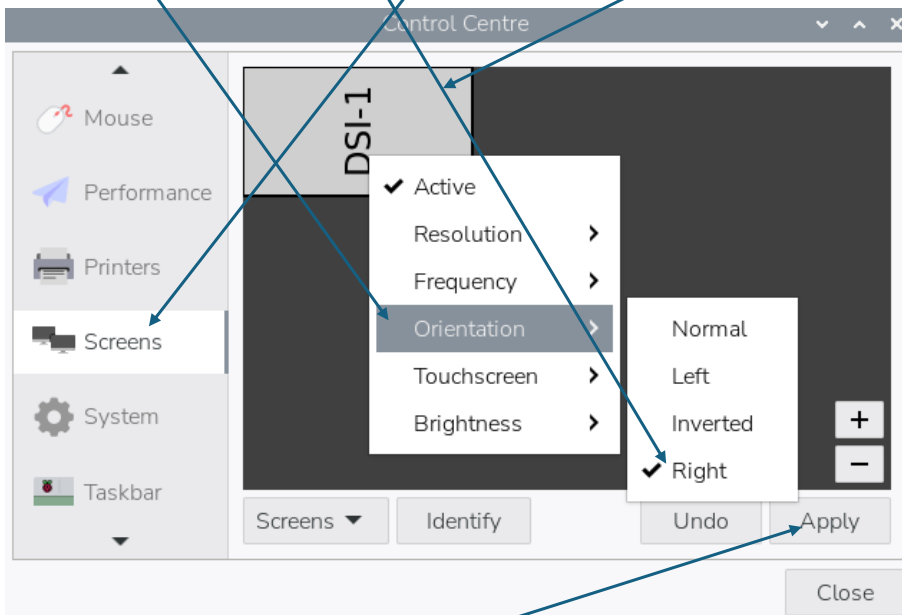
6. Once you edited the settings to your needs you can go back and start writing the OS on the SD card.
7. Only remove the SD card after getting a alert that tells you that the OS is finished flashing on the SD.

Screen

1. Once you connected the screen to the Raspberry Pi you can start it (this can take a few minutes, then the screen will turn on)
2. Now to rotate the screen horizontally you have to go to the top left and click the Raspberry Pi logo, then to preferences and then to control centre.



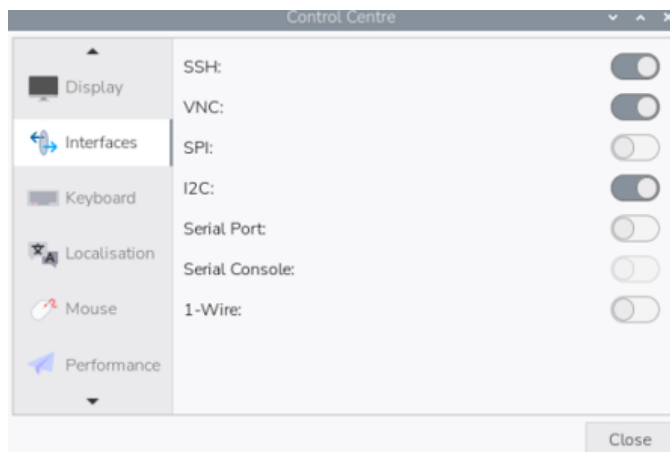
3. On the left scroll down to the Screens tab, then right-click the monitor, choose Orientation and select Right.



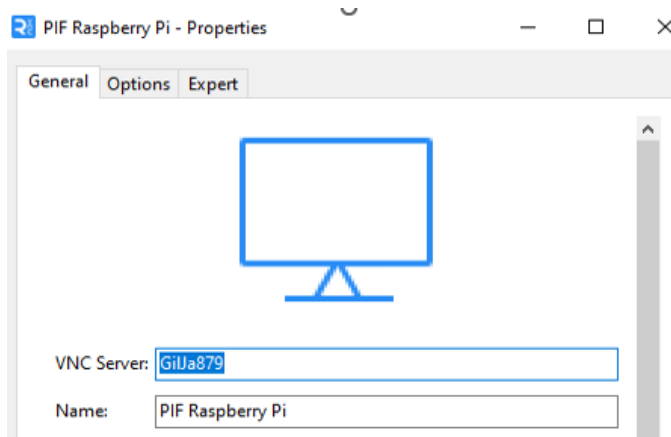
4. Press Apply and the screen will be horizontal (if you already connected your VNC viewer you will have to restart the connection to see the rotation)

SSH / I2C

1. To connect to your Raspberry Pi via VNC viewer you have to first enable SSH, this is also done in the control centre.
2. Navigate to the Interfaces tab and enable SSH, I2C and VNC.



3. Now start your VNC viewer and create a new connection, as VNC Server you can either type your hostname or your IP address and the Name can be whatever you want it's just a name that will show in your connections.



Static IP address

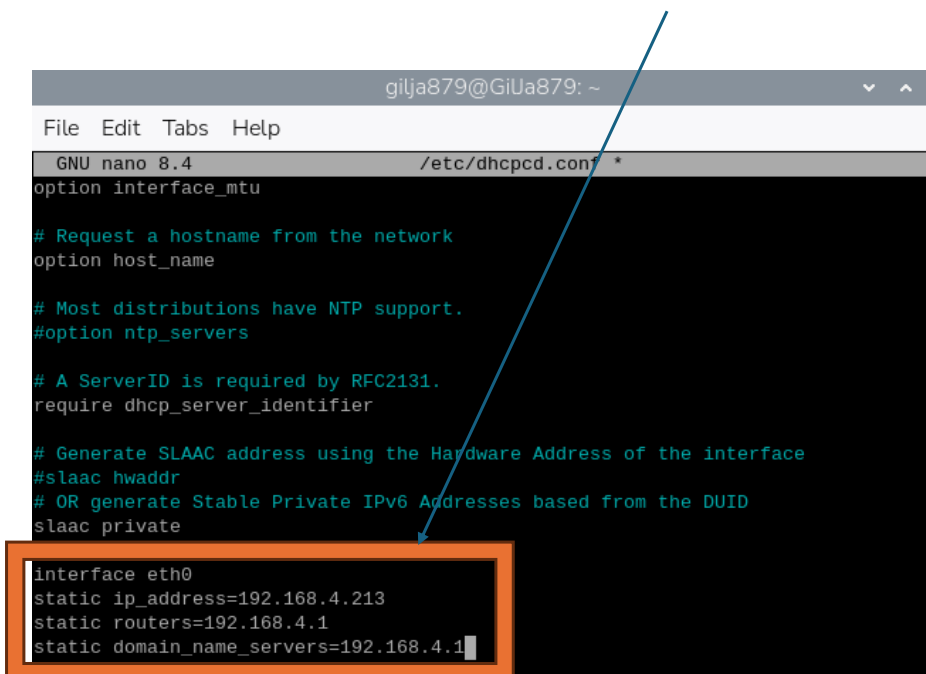
1. To set a static IP address we need to install dhcp, open your terminal and type the following command to install dhcp.

```
gilja879@GilJa879:~ $ sudo apt install dhcpd5
```

2. Once dhcp is installed you can open the dhcpd.conf file by the following command.

```
gilja879@GilJa879:~ $ sudo nano /etc/dhcpd.conf
```

3. In the file you will go to the bottom and add the lines under slaac private.



```
gilja879@GiUa879: ~
File Edit Tabs Help
GNU nano 8.4 /etc/dhcpd.conf *
option interface_mtu

# Request a hostname from the network
option host_name

# Most distributions have NTP support.
#option ntp_servers

# A ServerID is required by RFC2131.
require dhcp_server_identifier

# Generate SLAAC address using the Hardware Address of the interface
#slaac hwaddr
# OR generate Stable Private IPv6 Addresses based from the DUID
slaac private

interface eth0
static ip_address=192.168.4.213
static routers=192.168.4.1
static domain_name_servers=192.168.4.1
```

the first line tells dhcp which interface to use, the second line is the static ip address you want to use, the third line is the routers ip address and in our case the fourth line will also be the router ip address.

4. Once you edited the dhcpd.conf file you have to enable dhcp with the following command:

```
gilja879@GilJa879:~ $ sudo systemctl enable dhcpd
```

5. Now start your dhcp with the following command:

```
encodage: utf-8 --> utf-8
gilja879@GilJa879:~$ sudo systemctl start dhcpcd
```

Docker

1. Install docker with the following commands (write them one after the other in your terminal) :

```
gilja879@GilJa879:~$ sudo curl -fsSL https://get.docker.com -o install-docker.sh
```

```
gilja879@GilJa879:~$ sudo cat install-docker.sh
```

```
gilja879@GilJa879:~$ sudo sh install-docker.sh --dry-run
```

```
gilja879@GilJa879:~$ sudo sh install-docker.sh
```

2. Once that is done create a new directory where you will store the files given to you.

```
gilja879@GilJa879:~$ sudo mkdir /opt/station
gilja879@GilJa879:~$ cd /opt/station
gilja879@GilJa879:/opt/station$ sudo mkdir www
```

3. Download the files given to you.



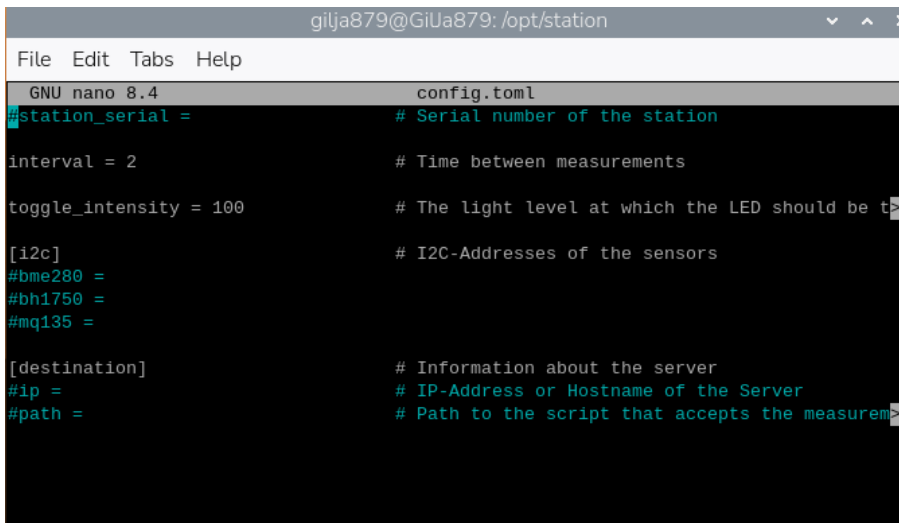
4. Navigate to the desktop and move the downloaded files into the directories you just create as shown below:

```
gilja879@GilJa879:~ $ cd Desktop/
gilja879@GilJa879:~/Desktop $ ls
'Firmware und Scripts'  Station-Webseite
gilja879@GilJa879:~/Desktop/Station-Webseite $ cd ..
gilja879@GilJa879:~/Desktop $ cd Firmware\ und\ Scripts/ .tation/www
gilja879@GilJa879:~/Desktop/Firmware und Scripts $ sudo mv * /opt/station
```

5. Now navigate to /opt/station and open the config.toml file to check if everything went correctly with the following command:

```
gilja879@GilJa879:~/Desktop/Firmware und Scripts $ cd /opt/station
gilja879@GilJa879:/opt/station $ sudo nano config.toml
```

The file should look like this:



```

gilja879@GilJa879: /opt/station
File Edit Tabs Help
GNU nano 8.4 config.toml
#station_serial = # Serial number of the station

interval = 2 # Time between measurements

toggle_intensity = 100 # The light level at which the LED should be t

[i2c] # I2C-Addresses of the sensors
#bme280 =
#bh1750 =
#mq135 =

[destination] # Information about the server
#ip = # IP-Address or Hostname of the Server
#path = # Path to the script that accepts the measurem

```

6. Now to start docker write the following command in your terminal:

```
gilja879@GilJa879:~ $ sudo docker run -d --name php --restart unless-stopped -p 8080:80 -v /tmp/station:/tmp/station -v /opt/station/www:/var/www/html php:8.4-apache
```

Show website in Kiosk mode

1. Install wtype with the following command:

```
gilja879@GiUa879:~ $ sudo apt -y install wtype
```

2. Enter the following command to edit the autostart file:

```
gilja879@GiUa879:~ $ sudo nano ~/.config/labwc/autostart
```

3. In the file copy the line below:

```
gilja879@GiUa879: /opt
GNU nano 8.4 /home/gilja879/.config/labwc/autostart *
chromium http://localhost:8080 --kiosk --noerrdialogs --disable-infobars --/switchtab.sh
```

This will start kiosk mode when the Raspberry Pi starts and show the correct website.

4. Open the switchtab.sh file:

```
gilja879@GiUa879:~ $ nano ~/switchtab.sh
```

5. In here copy the code that was provided to you:

```
gilja879@GiUa879: ~
File Edit Tabs Help
GNU nano 8.4 /home/gilja879/switchtab.sh *
#!/bin/bash

# Find Chromium browser process ID
chromium_pid=$(pgrep chromium | head -1)

# Check if Chromium is running
while [[ -z $chromium_pid ]]; do
    echo "Chromium browser is not running yet."
    sleep 5
    chromium_pid=$(pgrep chromium | head -1)
done

echo "Chromium browser process ID: $chromium_pid"

# Loop to send keyboard events
while true; do
    # Send Ctrl+Tab using `wtype` command
    wtype -M ctrl -P Tab -p Tab
    sleep 10
done
```

6. Now give yourself execute rights for the switchtab.sh file:

```
gilja879@GilJa879:~ $ chmod +x ~/switchtab.sh
```

7. Reboot the Raspberry Pi:

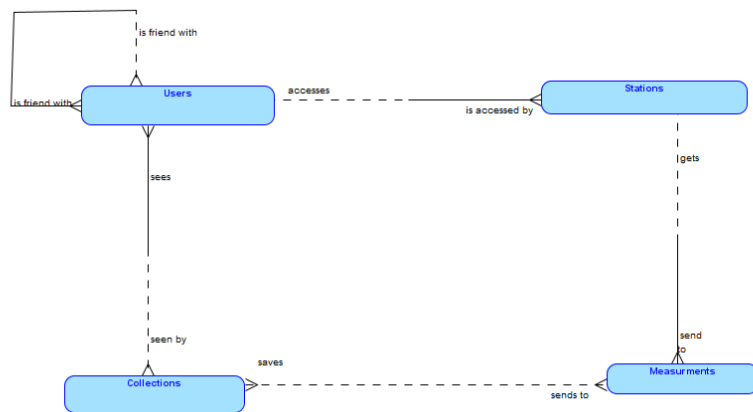
```
gilja879@GilJa879:~ $ sudo reboot
```

8. After the reboot, wait a few seconds and your screen should then show this website of the measurments:

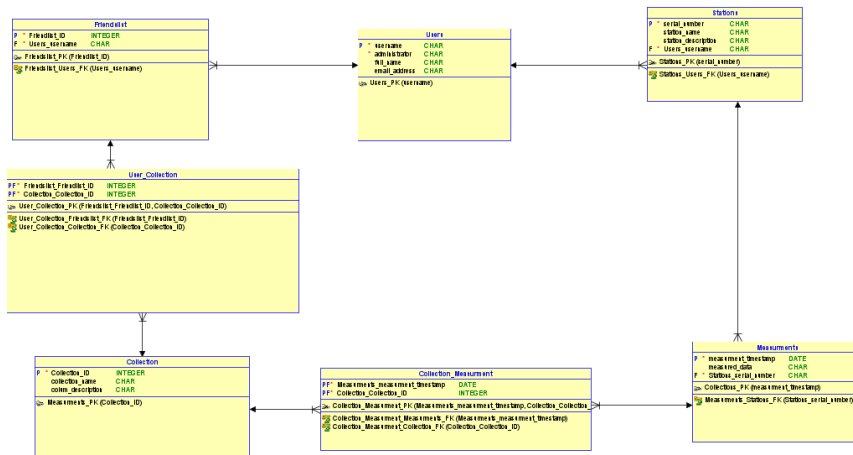
```
Warning: file(/tmp/station/result.csv): Failed to open stream: No such file or directory in /var/www/html/index.php on line 3
Time
Temperature      0.00 °C
Humidity          0.00 %
Pressure          0.00 hPa
Light             0.00 lux
Gas               0.00 ppm
```

L2 Database

Conceptual model



Logical/Physical Model



L3 Website

User Story	L3 - Website	to	KW 05
Informal description	As a user or administrator of a station		
	... I want a functional and user-friendly website		
	... so that I can manage and share stations and measured values.		
Requirements according to the work order	Required • Documentation: ○ Annotated source code of the website ○ Planning this user story Without login: • Login with username and password is possible. • Registration/creation of a new user account is possible. Logged-in users: • Logout is possible. • A simple welcome page is displayed. • Your own stations can be listed. The name and description of the stations can be changed. • Stations not yet assigned to a user can be registered by serial number. • Your own user account (username, password, email, first and last name) can be edited. • Measurement data from your own stations can be viewed and filtered by date/time. • Users can be added as friends. Friends can be displayed and friendships can be ended. • Measurement data can be added to collections. Users can only select from their own stations. A collection contains measurement data from a specific station from a start date/time to an end date/time. Your own collections can be renamed and deleted. • Collections can be shared with friends. Sharing can be undone. • Collections shared with the user can be viewed. Admins (additional): • All user accounts can be edited and deleted. Admin rights can be		

	assigned to users. New user accounts can be created. • All stations can be edited and deleted. New stations can be created. • Measurement data from all stations can be viewed, filtered by date/time, and deleted. • Measurement data from all stations can be added to collections. All collections can be renamed and deleted. Admins can only share collections they have created themselves. Should • The welcome page contains a dashboard that displays current measurement data from your own stations. • The website has an appealing and user-friendly design. • User entries are validated. • Users cannot be added as friends directly, but via a friend request system that requires confirmation of the request. • When a friendship <u>is ended, all collections</u> shared between the two users are automatically unshared. Could • Registration of stations via QR code. • Measurement data is displayed using diagrams. • The website automatically adapts to different screen sizes (responsive). • Different themes (e.g., light/dark mode). The selected theme is retained even after logging out. • The website is available in different languages and the selected language remains unchanged after logging out. • Users can be notified of various events (friend requests, sharing of measurement data, errors, etc.) via email and/or notifications. • Users can chat with each other on the website. • Users can send invite links to invite new users to the platform
Tasks according to work order • Create register page and connect it to the database. • Create login page according to the database users. • Create a homepage. • Create common <u>php file</u> to write shared code for all sites and reduce redundancy on each site. • Create a navigation bar to navigate between all sites in the common <u>php file</u>.	

- Create logout page and let user log out.
- Create account page to let users edit account information.
- Create station page, enables the user to add and edit their stations.
- Create collection page which allows user to see and create collections.
- Create Friending system that enables user to add or remove each other as friends.
- Create friendlist page to let the user see their friends and edit the list.

Resources

L4 Database Server

Apache installation

Apache is easily installed by just using this command:

```
gilja879@gilja879:~$ sudo apt install apache2
```

To check if Apache is running use the following command:

```
gilja789@gilja789-VirtualBox:~$ sudo systemctl status apache2
[sudo] password for gilja789:
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: v
   Active: active (running) since Wed 2026-03-11 08:08:30 CET; 57s ago
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 1149 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SU
   Main PID: 1198 (apache2)
    Tasks: 55 (limit: 19091)
   Memory: 7.9M (peak: 8.5M)
      CPU: 128ms
   CGroup: /system.slice/apache2.service
           └─1198 /usr/sbin/apache2 -k start
             └─1200 /usr/sbin/apache2 -k start
               └─1201 /usr/sbin/apache2 -k start
```

Mariadb installation

Type the following command to install maria db

```
gilja789@gilja789-VirtualBox:~$ sudo apt install mariadb-server mariadb-client -y
```

Then start mariadb with `sudo systemctl start mariadb`.

To check if it works use `sudo systemctl status mariadb`.

```
gilja789@gilja789-VirtualBox:~$ sudo systemctl start mariadb
gilja789@gilja789-VirtualBox:~$ sudo systemctl status mariadb
● mariadb.service - MariaDB 10.11.14 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: >
   Active: active (running) since Wed 2026-03-11 08:12:19 CET; 1min 14s ago
     Docs: man:mariadb(8)
```

SQL

To start a sql line you write `sudo mysql`, since your mariadb is running it will activate in your sql line.

The commands you must use are marked by orange boxes.

1. First you create a database (in my case I create a test database, but when you already want to add your website create a database with the same name as the one used in your website).
2. Then you create a sql user and give it a password.
3. Afterwards you give the user rights to your new database.
4. Finally you flush the privileges which saves the rights for your user.

```
gilja789@gilja789-VirtualBox:~$ sudo mysql
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 35
Server version: 10.11.14-MariaDB-0ubuntu0.24.04.1 Ubuntu 24.04

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> CREATE DATABASE testdb;
Query OK, 1 row affected (0.001 sec)

MariaDB [(none)]> CREATE USER 'gilja879'@'localhost' IDENTIFIED BY 'gilja879';
Query OK, 0 rows affected (0.013 sec)

MariaDB [(none)]> GRANT ALL PRIVILEGES ON testdb.* TO 'GilJa879'@'localhost';
Query OK, 0 rows affected (0.000 sec)

MariaDB [(none)]> GRANT ALL PRIVILEGES ON testdb.* TO 'gilja879'@'localhost';
Query OK, 0 rows affected (0.004 sec)

MariaDB [(none)]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.001 sec)
```

PHP installation

Copy the following command to get all php extensions you need for your apache.

```
gilja789@gilja789-VirtualBox:~$ sudo apt install php libapache2-mod-php php-cli
php-curl php-gd php-mbstring php-xml php-zip -y
```

Then restart your apache to make sure it uses the php.

```
gilja789@gilja789-VirtualBox:~$ sudo systemctl restart apache2
```

phpMyAdmin

First, we install a tool to unzip zip files.

```
gilja789@gilja789-VirtualBox:~$ sudo apt install unzip wget -y
```

Then copy this command to get the zip file for phpMyAdmin.

```
gilja789@gilja789-VirtualBox:~$ wget https://www.phpmyadmin.net/downloads/phpMyAdmin-latest-all-languages.zip
```

Now we use the unzip tool and unzip the phpMyAdmin zip file.

```
gilja789@gilja789-VirtualBox:~$ unzip phpMyAdmin-latest-all-languages.zip
```

Then move the unzipped files to /var/www/html so that you can access it via “your ip address/phpMyAdmin”.

```
gilja789@gilja789-VirtualBox:~$ sudo mv phpMyAdmin-*all-languages /var/www/html/phpmyadmin
```

Finally give your sql user rights for the directory so you can login with your sql user in phpMyAdmin.

```
gilja789@gilja789-VirtualBox:~$ sudo chown -R www-data:www-data /var/www/html/phpmyadmin
```

```
gilja789@gilja789-VirtualBox:~$ sudo chown -R www-data:www-data /var/www/html
gilja789@gilja789-VirtualBox:~$ sudo chmod -R 755 /var/www/html
```

FileZilla

First we need to install ssh so that we can access the vm from the host machine.

!!! Make sure to have bridged adapter and not NAT in your network settings!!!

```
gilja789@gilja789-VirtualBox:~$ sudo apt install openssh-server -y
```

Once installed we will now start ssh and enable it so that it will always automatically start.

```
gilja789@gilja789-VirtualBox:~$ sudo systemctl start ssh
gilja789@gilja789-VirtualBox:~$ sudo systemctl enable ssh
```

Use following commands to set up ssh.

```
gilja789@gilja789-VirtualBox:~$ sudo ufw allow OpenSSH
Rules updated
Rules updated (v6)
gilja789@gilja789-VirtualBox:~$ sudo ufw reload
Firewall not enabled (skipping reload)
```

```
gilja789@gilja789-VirtualBox:~$ sudo a2enmod ssl
```

Now on your host machine open Filezilla and enter the ip, username, password and the port (22).

Host: sftp://192.168.4.138 Username: gilja789 Password: ***** Port: Quickconnect

Error: File transfer failed
Status: Retrieving directory listing of "/var/www/html" ...
Status: Listing directory "/var/www/html"
Status: Directory listing of "/var/www/html" successful
Status: Disconnected from server
Status: Disconnected from server

Local site: C:\xampp\htdocs\WSERS-PHP-1\PIF

Remote site: /var/www/html

Filename	Filesize	Filetype	Last modified
Website		File folder	15.01.26 11:14:14
Account.php	7 621	PHP Source File	20.01.26 11:02:41
collections.php	9 780	PHP Source File	20.01.26 10:48:03
commonphp.php	6 018	PHP Source File	21.01.26 09:27:50
Friendlist.php	465	PHP Source File	14.01.26 09:49:53
Homepage.php	457	PHP Source File	14.01.26 09:49:53
index.php	2 534	PHP Source File	20.01.26 10:48:03
L2 Database.sql	2 857	SQL Source File	22.01.26 11:42:37
Logout.php	1 135	PHP Source File	14.01.26 09:49:57
Measurments.php	341	PHP Source File	21.01.26 08:43:35
MyCss.css	1 907	Cascading Style Sh...	20.01.26 11:07:56
Register.php	5 221	PHP Source File	20.01.26 10:48:03
Stations.php	9 413	PHP Source File	22.01.26 11:43:23
translation.csv	3 108	Microsoft Excel Co...	15.01.26 11:14:14

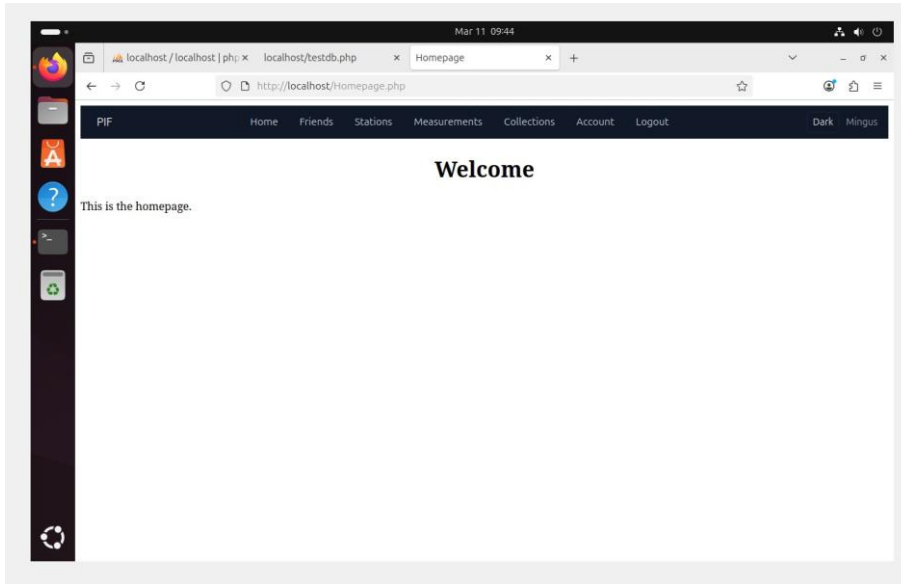
13 files and 1 directory. Total size: 50 857 bytes

Filename	Filesize	Filetype	Last modified	Permissions
phpmyadmin		File folder	11.03.26 08:29:10	drwxr-xr-x
Account.php	7 621	PHP Sourc...	11.03.26 09:13:01	-rw-rw-r--
collections.php	9 780	PHP Sourc...	11.03.26 09:13:01	-rw-rw-r--
commonphp.php	6 018	PHP Sourc...	11.03.26 09:13:01	-rw-rw-r--
Friendlist.php	465	PHP Sourc...	11.03.26 09:13:01	-rw-rw-r--
Homepage.php	457	PHP Sourc...	11.03.26 09:13:01	-rw-rw-r--
index.html	10 671	Chrome H...	05.03.26 11:39:39	-rwxr-xr-x
index.php	20	PHP Sourc...	11.03.26 08:10:49	-rwxr-xr-x
L2 Database.sql	2 857	SQL Sourc...	11.03.26 09:13:01	-rw-rw-r--
Logout.php	1 135	PHP Sourc...	11.03.26 09:13:01	-rw-rw-r--
Measurments.php	341	PHP Sourc...	11.03.26 09:13:01	-rw-rw-r--
MyCss.css	1 907	Cascading ...	11.03.26 09:13:01	-rw-rw-r--
Register.php	5 221	PHP Sourc...	11.03.26 09:13:01	-rw-rw-r--
Stations.php	9 413	PHP Sourc...	11.03.26 09:13:01	-rw-rw-r--
testdb.php	195	PHP Sourc...	11.03.26 08:39:39	-rw-r--r--
translation.csv	3 108	Microsoft ...	11.03.26 09:13:02	-rw-rw-r--

15 files and 1 directory. Total size: 59 209 bytes

Once you moved your website into the /var/www/html directory you should be able to access it from your vm.

If you have a database make sure to change the mysqli_connection variables to your new username and password (your sql user).

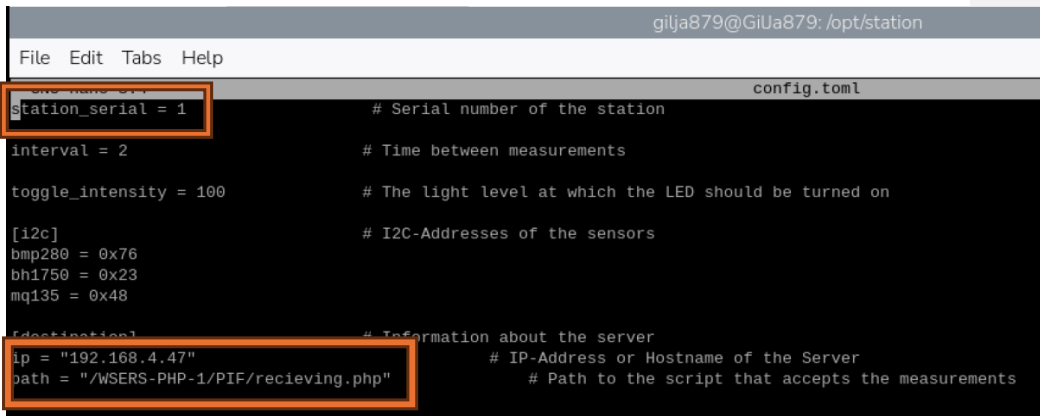


L5 Data Transfer

Config.toml

I made some changes in my config.toml file.

- station_serial = 1 | Changed the station serial number to the number of my station.
- Ip = "192.168.4.47" | Changed the ip address to the ip address of my server (VM).
- Path = "/WSERS-PHP-1/PIF/recieving.php" | Changed the path to the path of the newly created php file that recieves the data from the station.



```
gilja879@GiUa879: /opt/station
File Edit Tabs Help
config.toml
station_serial = 1          # Serial number of the station
interval = 2               # Time between measurements
toggle_intensity = 100     # The light level at which the LED should be turned on
[i2c]                     # I2C-Addresses of the sensors
bmp280 = 0x76
bh1750 = 0x23
mq135 = 0x48
[destination]             # Information about the server
ip = "192.168.4.47"       # IP-Address or Hostname of the Server
path = "/WSERS-PHP-1/PIF/recieving.php" # Path to the script that accepts the measurements
```

Receiving.php

```
receiving.php > ...
<?php
include_once("commonphp.php");

// require POST
if ($_SERVER['REQUEST_METHOD'] != 'POST') {
    http_response_code(405);
    echo 'Method not allowed';
    exit;
}

// minimal required fields
$required = ['timestamp','temperature','humidity','pressure','light','gas','station_serial'];
foreach ($required as $f) {
    if (!isset($_POST[$f])) {
        http_response_code(400);
        echo "Missing field: $f";
        exit;
    }
}

$timestamp = $_POST['timestamp'];
$temperature = (int)$_POST['temperature'];
$humidity = (int)$_POST['humidity'];
$pressure = (int)$_POST['pressure'];
$light = (int)$_POST['light'];
$gas = (int)$_POST['gas'];
$station = (int)$_POST['station_serial'];

// prepare statement
$stmt = $conn->prepare(
    "INSERT INTO 'Measurement' (timestamp,Measurement, temperature, humidity, airpressure, lightintensity, airquality, station) VALUES (?,?,,?,?,,?)"
);
if (!$stmt) {
    http_response_code(500);
    echo "Prepare failed: " . $conn->error;
    exit;
}

// bind and execute
if (!$stmt->bind_param("siiiiii", $timestamp, $temperature, $humidity, $pressure, $light, $gas, $station)) {
    http_response_code(500);
    echo "Bind failed: " . $stmt->error;
    $stmt->close();
    exit;
}

if (!$stmt->execute()) {
    http_response_code(500);
    echo "Execute failed: " . $stmt->error;
    $stmt->close();
    exit;
}

$stmt->close();
http_response_code(201);
echo "OK";
?>
```

This script will get the measurments using the POST method and then write the data into the database.

Proof

To show that the measurements are being sent to the database, I simply ran the measurement table and checked if measurements are being added.

I attached a video of the more data being added as proof.

Problem/Solution

Since my website cannot be accessed without being logged in, the data could not get sent to the receiving.php, I had to change my commonphp code to allow the access to the receiving.php so that the measurements can be put in the database.

```
32
33 // Simple access control: allow only index.php and Register.php for anonymous users
34 $publicPages = ['index.php', 'Register.php', 'receiving.php'];
35 $currentPage = basename($_SERVER['PHP_SELF']);
36
```

I simply added the receiving.php to the sites that are accessible without being logged in.

L6 Backup Server

Backup VM

First install mysql on the new vm.

```
gilja879@gilja879:~$ sudo apt install mysql-client -y
```

Then create a directory where you will later save the backups.

```
gilja879@gilja879:~$ sudo mkdir -p /backup//database
```

Create a new user and give him rights for the backup directory.

```
gilja879@gilja879:~$ sudo useradd -m backupuser
gilja879@gilja879:~$ sudo passwd backupuser
New password:
Retype new password:
passwd: password updated successfully
gilja879@gilja879:~$ sudo chown backupuser:backupuser /backup/database
```

Create a sh file in /usr/local/bin. This script will create the backup.

```
gilja879@gilja879:/usr/local/bin$ sudo nano backup.sh
```

```
GNU nano 7.2 backup.sh
#!/bin/bash

#Variables
DB_HOST="<192.168.4.138>"
DB_USER="backupuser"
DB_NAME="PIF"
BACKUP_DIR="/backup/database"
DATE=$(date +%Y-%m-%d_%H-%M-%S)
BACKUP_FILE="$BACKUP_DIR/${PIF}_${DATE}.sql"

#Run the backup
mysqldump -h $DB_HOST -u $DB_USER <PIF> > "$BACKUP_FILE"
```

Make the sh file executable.

```
gilja879@gilja879:/usr/local/bin$ sudo chmod +x backup.sh
gilja879@gilja879:/usr/local/bin$
```

Create the crontab to schedule the backups.

```
gilja879@gilja879:/usr/local/bin$ crontab -e
```

Change this line. This will run the sh file every day of every month, all year at 8 AM.

```

GNU nano 7.2 /tmp/crontab.AKaVog/crontab
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 1 a.m every week with:
# 0 1 * * * /usr/local/bin/backup.sh
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow   command
  
```

Now set up FTP access, first install vsftpd.

```

gilja879@gilja879:~$ sudo apt install vsftpd -y
  
```

Create a ftp user.

```

gilja879@gilja879:~$ sudo useradd -m ftpuser
gilja879@gilja879:~$ sudo passwd ftpuser
New password:
Retype new password:
passwd: password updated successfully
  
```

Commented [JG1]: Password 1

Give the ftp user rights.

```

gilja879@gilja879:~$ sudo chown ftpuser:ftpuser /backup/database
  
```

Enter the vsftpd.conf

And uncommand write_enable=YES and local_enable=YES

```

gilja879@gilja879:~$ sudo nano /etc/vsftpd.conf
  
```

```
GNU nano 7.2 /etc/vsftpd.conf *
# Example config file /etc/vsftpd.conf
#
# The default compiled in settings are fairly paranoid. This sample file
# loosens things up a bit, to make the ftp daemon more usable.
# Please see vsftpd.conf.5 for all compiled in defaults.
#
# READ THIS: This example file is NOT an exhaustive list of vsftpd options.
# Please read the vsftpd.conf.5 manual page to get a full idea of vsftpd's
# capabilities.
#
#
# Run standalone? vsftpd can run either from an inetd or as a standalone
# daemon started from an initscript.
listen=NO
#
# This directive enables listening on IPv6 sockets. By default, listening
# on the IPv6 "any" address (:::) will accept connections from both IPv6
# and IPv4 clients. It is not necessary to listen on *both* IPv4 and IPv6
# sockets. If you want that (perhaps because you want to listen on specific
# addresses) then you must run two copies of vsftpd with two configuration
# files.
listen_ipv6=YES
#
# Allow anonymous FTP? (Disabled by default).
anonymous_enable=NO
#
# Uncomment this to allow local users to log in.
local_enable=YES
#
# Uncomment this to enable any form of FTP write command.
write_enable=YES
#
```

Server VM

Enter mysql as root user.

```
gilja789@gilja789-VirtualBox:~$ sudo mysql -u root -p
```

Create the user from the backup server in your database.

```
MariaDB [PIF]> CREATE USER 'backupuser'@'192.168.4.100' IDENTIFIED BY '1';
```

And give him permission for the database.

```
MariaDB [PIF]> GRANT ALL PRIVILEGES ON PIF.* TO 'backupuser'@'192.168.4.100';
Query OK, 0 rows affected (0.004 sec)
```

```
MariaDB [PIF]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.001 sec)
```

Now allow MYSQL to accept external connections, to be able to connect to the backup vm.

```
gilja789@gilja789-VirtualBox:/etc/mysql/mariadb.conf.d$ sudo nano 50-server.cnf
```

Change the bind address to 0.0.0.0 to allow all ip addresses.

```

gilja789@gilja789-VirtualBox: /etc/mysql/mariadb.conf.d
GNU nano 7.2 50-server.cnf
#user                        = mysql
pid-file                    = /run/mysqld/mysqld.pid
basedir                    = /usr
#datadir                    = /var/lib/mysql
#tmpdir                     = /tmp

# Broken reverse DNS slows down connections considerably and name resolu
# to skip if there are no "host by domain name" access grants
#skip-name-resolve

# Instead of skip-networking the default is now to listen only on
# localhost which is more compatible and is not less secure.
bind-address                = 0.0.0.0

#
# * Fine Tuning
  
```

Finally back on the backup vm, restart vsftpd.

```

gilja879@gilja879:~$ sudo systemctl restart vsftpd
gilja879@gilja879:~$ sudo systemctl enable vsftpd
  
```

Run the sh file to test if the backup is being saved.

```

gilja879@gilja879:/usr/local/bin$ sudo sh backup.sh
  
```

Go to the backup location to check.

```

gilja879@gilja879:~$ cd /backup/database/
gilja879@gilja879:/backup/database$ ls
_2026-04-16_13-23-42.sql  _2026-04-16_13-32-23.sql
gilja879@gilja879:/backup/database$
  
```

Workshop sprints

Userstory A1

Scraping / Punching

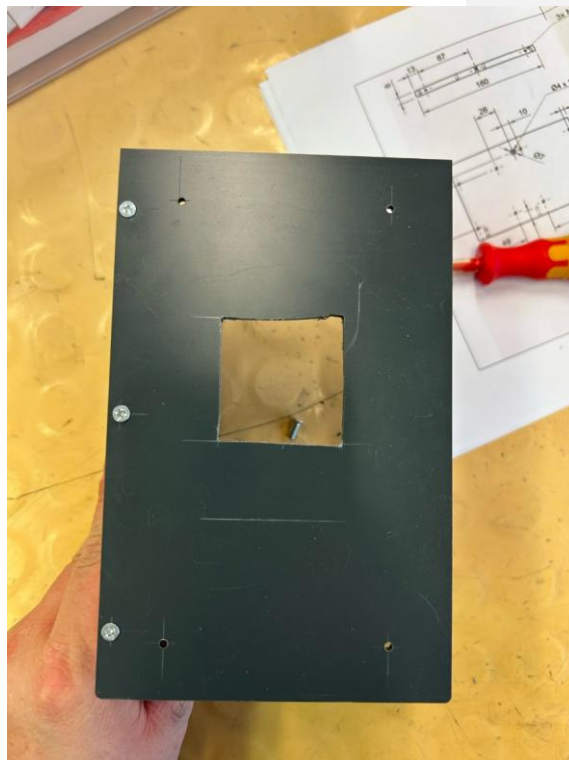
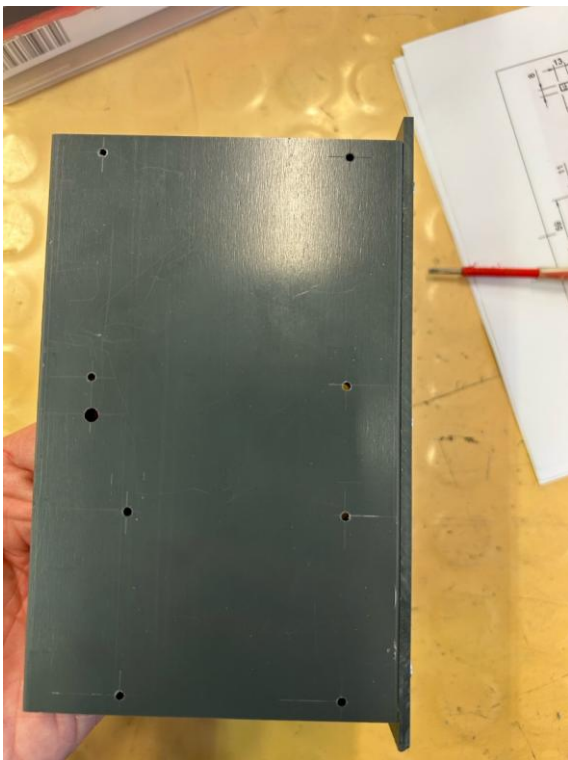
First, I scraped lines according to the measurements of the holes and then punched where the hole is supposed to be, that helps as a guiding hole for the drill.



Drilling

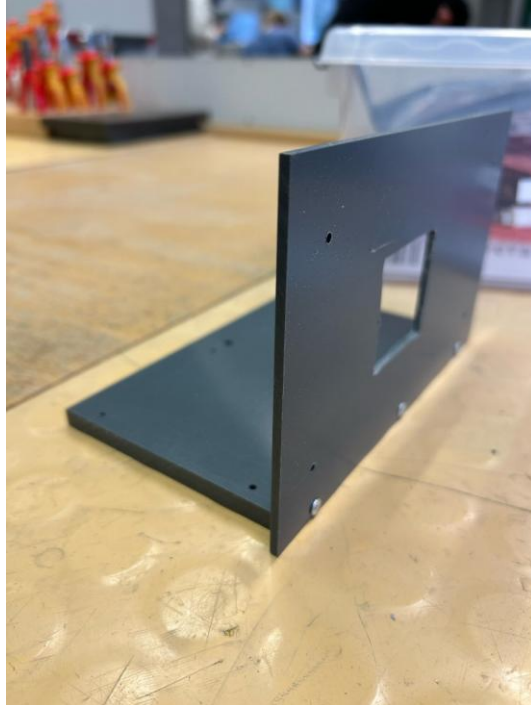
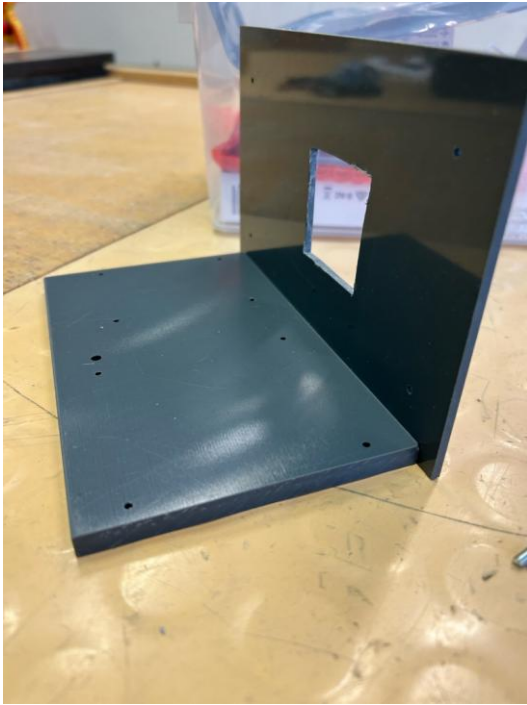
I then drilled out the holes, the holes on the side had to be drilled with a special drill so that the screws can be screwed in.

And for the square in the middle of the plate I made 4 big holes on the edges and then sawed out the rest of the square.



Screwing together

I then screwed the two plates together.



Installing the display and raspberry pi

Now I screwed in the display on the front plate in the holes I drilled earlier, and the square is there to put the cables trough to connect the display to the raspberry pi.



To screw in the raspberry pi, we used spacers so that the bottom of the raspberry pi can't be damaged.



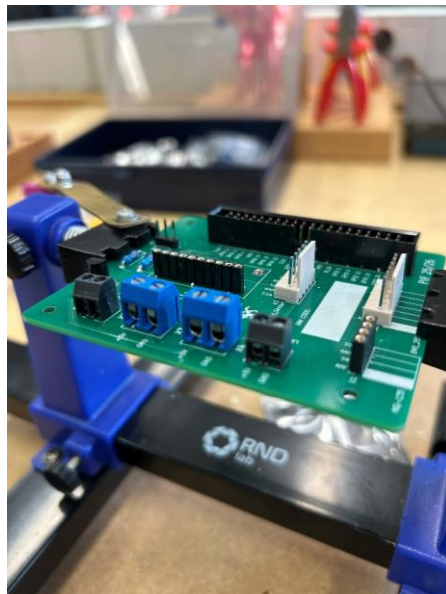
Userstory A2

Soldering

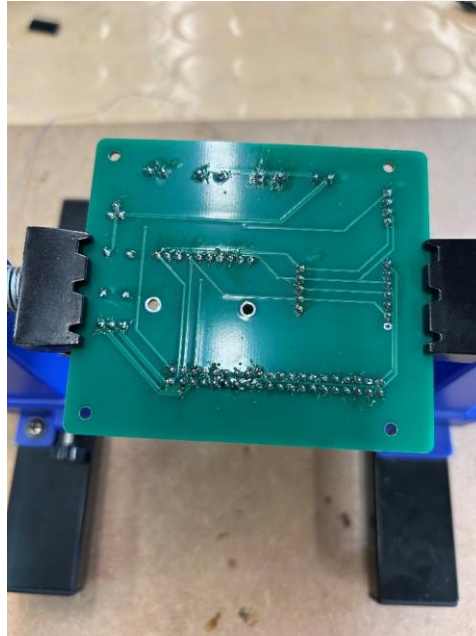
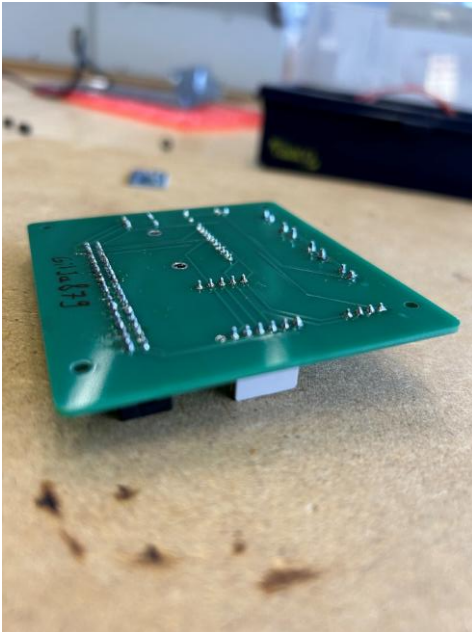
First, I put my circuit board in this holder so it's easier to solder everything on the board, it makes it more accessible.



I Then soldered every component onto the circuit board; I started with the smallest pieces to make it easier to reach.



On the underside I cut off the parts that were too long with a flush wire cutter, so it will fit and nothing can get caught on it.



The finished circuit board:



Soldering station:

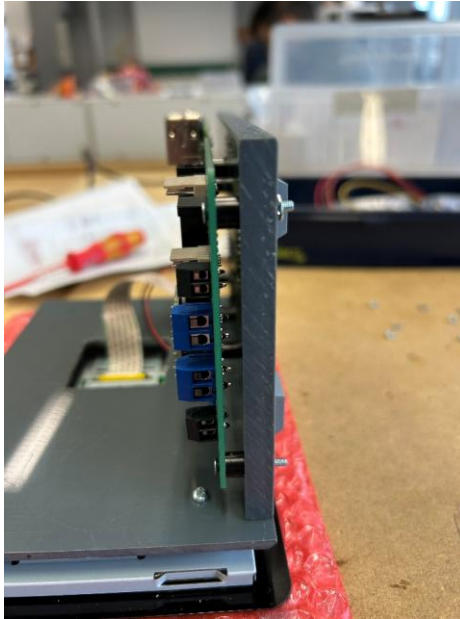


Put circuit board on the pvc plate.

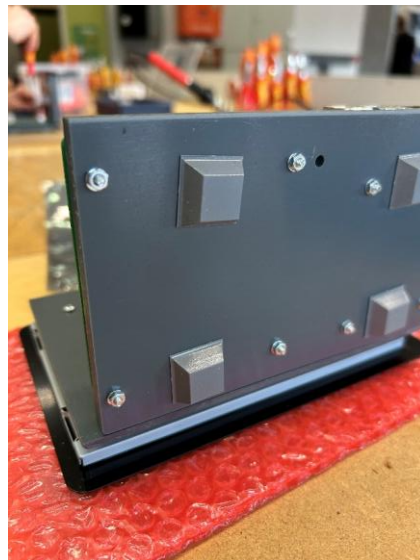
There are 4 screws that will hold the circuit board down on each edge.



Between the circuit board and the pvc plate there will be 4 spacers, where the screw goes through so that the circuit board can't get damaged.



On the back to hold the screw in place, I put a washer on each screw and then screwed in a nut on each screw.



Userstory A3

Cutting the cable in half



Removing the outer jacket

I removed the first jacket with a wire stripper and pulled back the shield to not damage it.



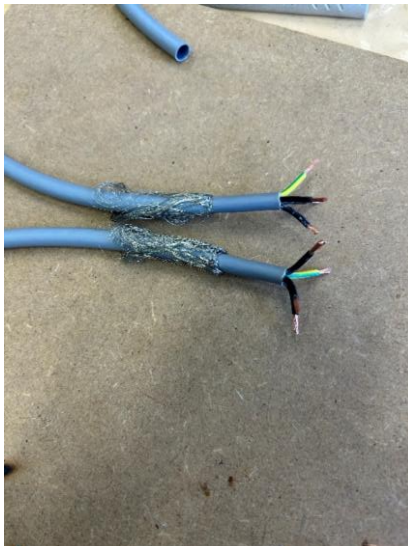
Removing the inner jacket and staggering

Then I removed the inner jacket to reveal the conductors. Afterwards I will put each conductor next to each other and cut in the middle so that all the conductors will be the same length.



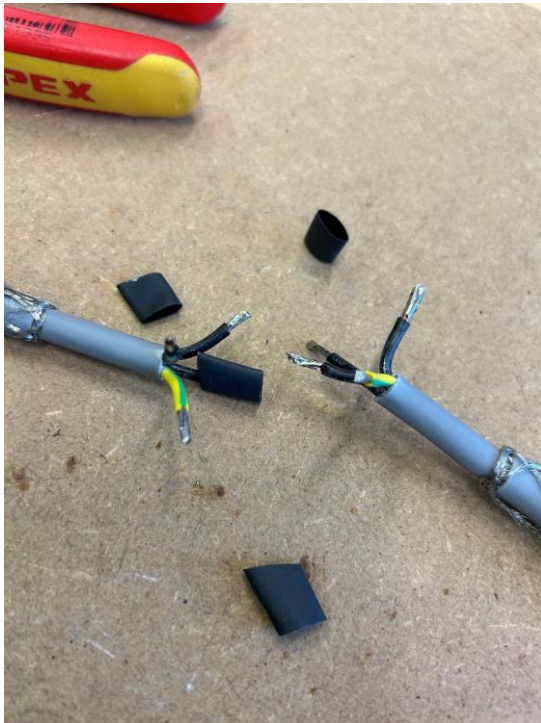
Removing the conductor shield.

After I removed the shield from the conductors with a wire stripper to reveal the copper, which allows me to connect the cables together.



Soldering

Then I soak the copper wires in solder, then I will just put both ends together and heat them up until the solder will connect both ends together.



Heat shrink tube

After the conductors are connected I put the heat shrink tube over the soldered spot and heat it up with a heat gun to make it shrink in position, protecting the soldered spot.



Userstory A4

16Bit Converter and gas module

First, I soldered the sensor onto the pins and then put them on the connectors on the circuit board.



To prove it worked, I checked with a command to see if the sensor can see the other sensors:

```

gilja879@GilJa879:/opt/station $ i2cdetect -y 1
    0  1  2  3  4  5  6  7  8  9  a  b  c  d  e  f
00: -- -- -- -- -- -- -- -- -- -- -- -- -- --
10: -- -- -- -- -- -- -- -- -- -- -- -- -- --
20: -- -- -- 23 -- -- -- -- -- -- -- -- -- --
30: -- -- -- -- -- -- -- -- -- -- -- -- -- --
40: -- -- -- -- -- -- -- 48 -- -- -- -- -- --
50: -- -- -- -- -- -- -- -- -- -- -- -- -- --
60: -- -- -- -- -- -- -- -- -- -- -- -- -- --
70: -- -- -- -- -- 76 -- -- -- -- -- -- -- --
  
```

Afterwards I put the gas module on the connectors on the raspberry pi.

To see if it works, I used a command to read the measurements of the gas module a few times to see if it changes (Result in the video attached).

Userstory A5

BHI750 sensor

First, I soldered the sensor to the wires we were provided with and simply plugged the connector of the wires onto my circuit board.

To test if the sensor gets any data, I ran the python test script to see if I get different results depending on how bright it is since this is a brightness sensor. (video attached).

BME280 sensor

Same as for the bhi750 sensor I first soldered the sensor to the provided cables and then simply plugged it onto my circuit board.

To test I also ran the python test file of this sensor to see if I get data. This sensor reads air pressure, temperature and humidity. (video attached).

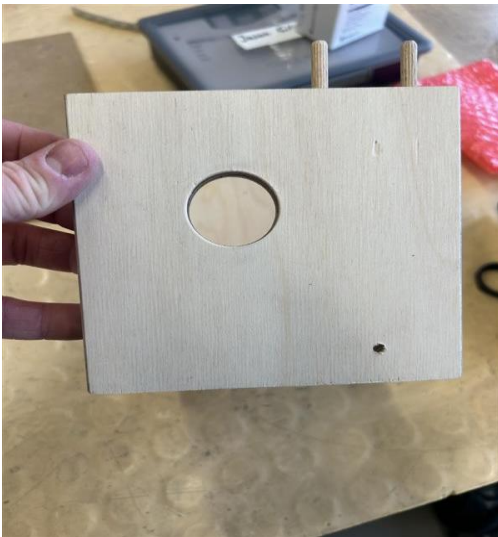
Problems

There was a problem with the script we were provided with, for the BME280 sensor so it took a long time to figure out how to get data. I found a problem was the library we used for the sensor and after some research I found a working library, but the sensor can now only show temperature and air pressure, not the humidity. I still have to fix this issue.

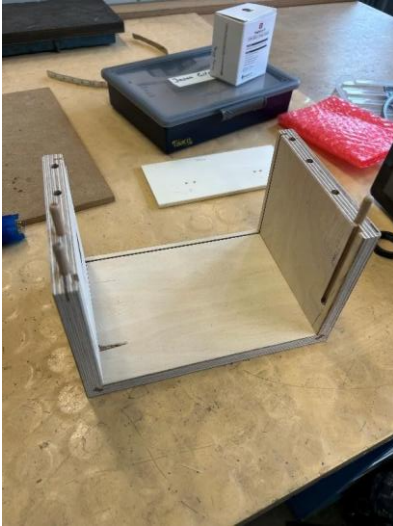
Userstory A6

Box Assembly

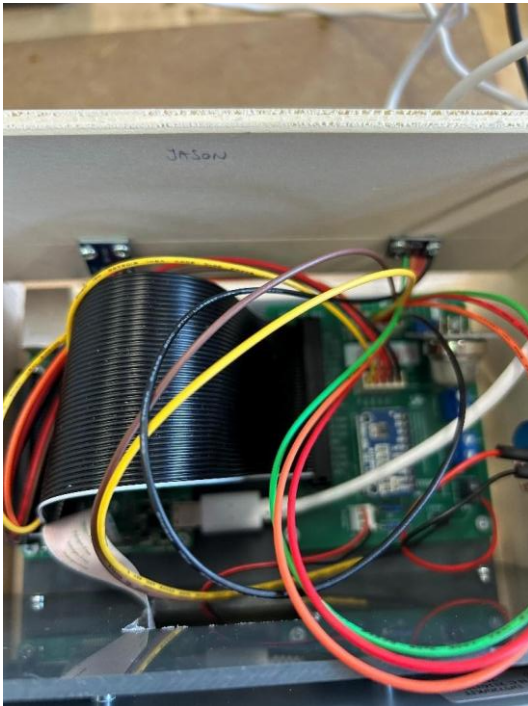
First, I prepared and drilled all the holes in the wooden plates.



And carved out the space for the screen and the backplate and put it all together.

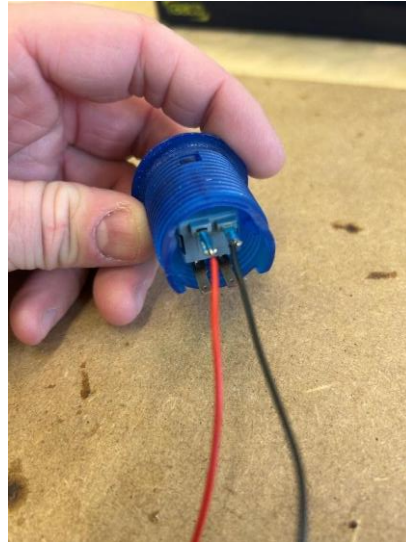


Finally, I put my raspberry pi in the box.

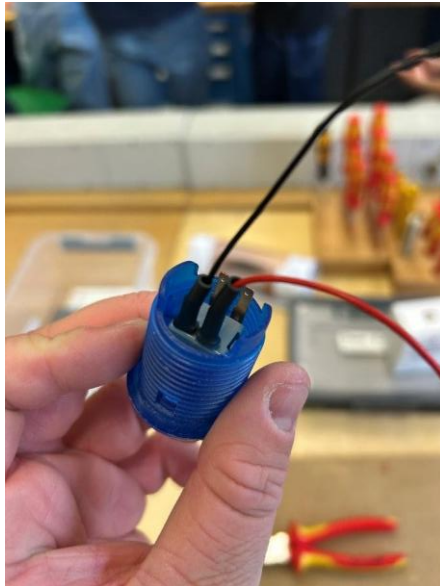


Button

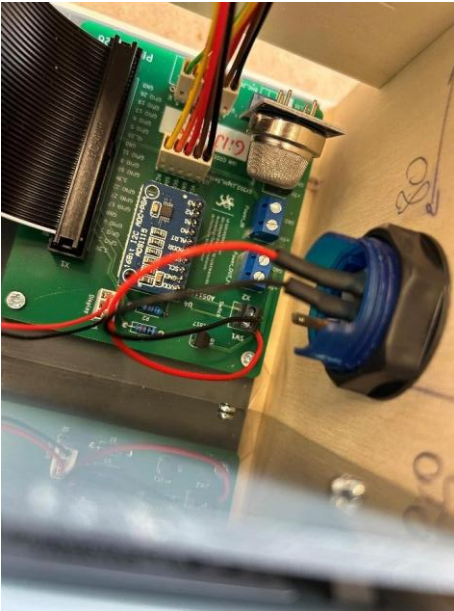
First, I removed the shield cover from the cables and soldered them to the button.



Then I added a heat shrink around the cables I just soldered to protect them.



Finally, I put the button in the wooden case and connected it to the circuit board connector.



Video of proof of it working is attached as separate file.

Material List

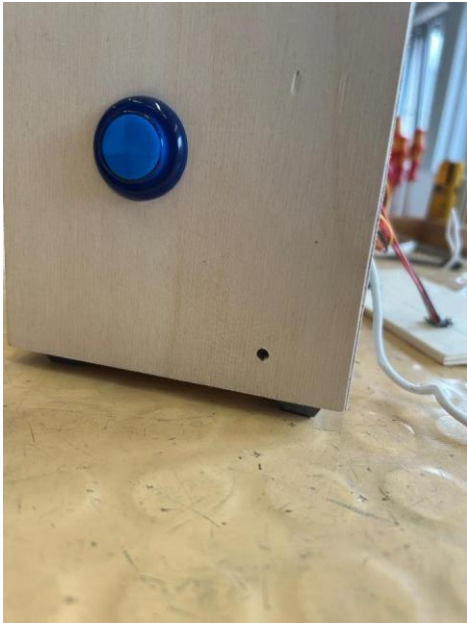
- Lead
- Soldering station
- Drill
- Screws
- Button
- Cables
- Screwdriver

Userstory A7

1. First, I prepared two cables, I removed the outer jacket on both sides so that I can solder it onto the LED strip and connect the other side to my circuit board.



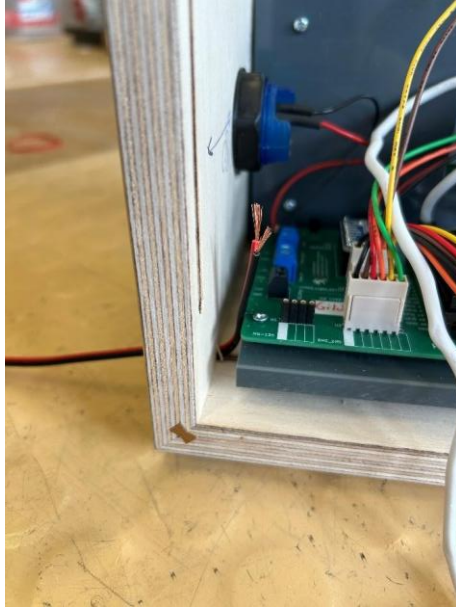
2. The cable will go through a hole on the side.



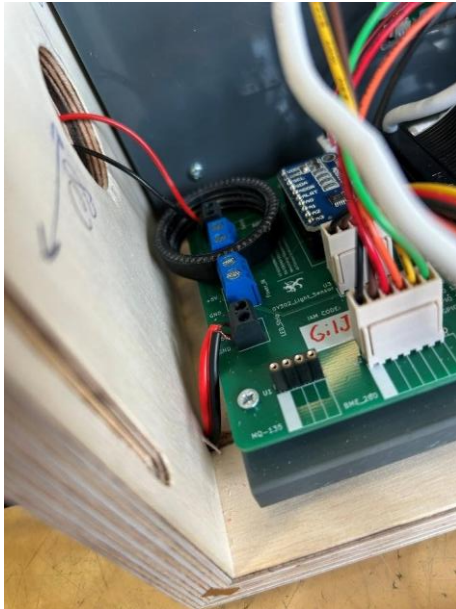
3. I soldered the cables to the LED strip.



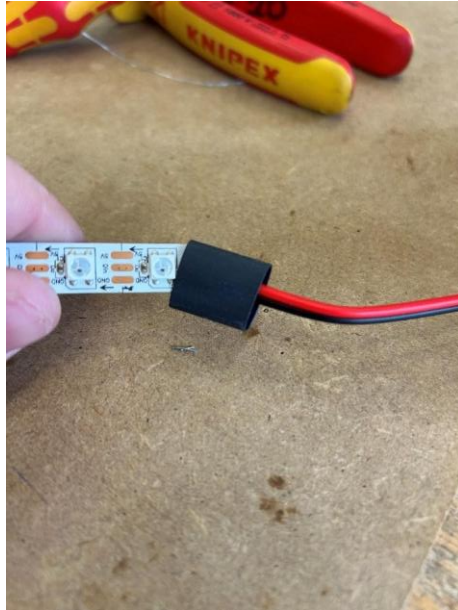
4. Then I pushed the other side of the cable trough the hole.



5. And connected the cables to the circuit board.



6. I added a heat shrink where I connected the cables to the LED strip to keep the connection save.



Experience

This task worked well and was easy to understand, however I made a small mistake by removing the jacket too short which made it harder to remove the shield from the conductors and soldering them together.

Equipment

- Shielded cable
- Diagonal cutter
- Soldering station
- Solder
- Heat shrink tube
- Wire stripper